

Software Requirements Specification ? ShopSphere E-Commerce Platform

1. Introduction

This document specifies the requirements for the ShopSphere e-commerce platform. It is intended for the development, quality assurance and operations teams.

2. Account Management

R001: The system shall allow new users to create an account using an email address and a password.

R002: The system shall allow registered users to log in using their email address and password.

R003: The system shall allow users to log in using their mobile number and password only.

R004: The system shall allow a user to reset a forgotten password through a password reset link.

R005: The system shall lock a user account after five consecutive failed login attempts.

R006: The system shall send notifications.

R007: The system shall allow users to update their profile information.

3. Product Catalogue

R008: The system shall display a catalogue of products with name, price, description and availability.

R009: The system shall allow customers to search for products by keyword.

R010: The system shall allow customers to filter search results by category, price range and rating.

R011: Search results shall be returned quickly.

R012: The system shall display product reviews submitted by verified purchasers.

R013: The system shall support at least 50,000 product listings.

4. Shopping Cart and Checkout

R014: The system shall allow a logged-in customer to add products to a shopping cart.

R015: The system shall allow a customer to remove products from the shopping cart.

R016: The system shall calculate the cart total including applicable taxes and shipping charges.

R017: The system shall allow customers to register for an account.

R018: The system shall allow customers to complete a purchase using the checkout process.

R019: Guest checkout is not allowed; all purchases require a registered account.

R020: The system shall allow users to check out without creating an account.

R021: The system shall support payment by credit card, debit card and digital wallet.

R022: The system shall send an order confirmation email after a successful purchase.

5. Order Management

R023: The system shall allow customers to view their order history.

R024: The system shall allow customers to track the delivery status of a placed order.

R025: The system shall allow customers to cancel an order before it has been shipped.

R026: The application shall respond quickly to user requests.

R027: The system shall allow administrators to update the status of an order.

R028: Refunds shall be processed in a reasonable time frame.

6. Administration

R029: The system shall allow administrators to add, edit and remove product listings.

R030: The system shall allow administrators to view sales reports.

R031: The system shall provide an appropriate level of access control for administrative functions.

R032: The system shall record an audit log of all administrative actions.

R033: The system shall allow administrators to manage user accounts.

7. Security

R034: The system shall encrypt all stored customer passwords.

R035: All data transmitted between the client and the server shall be encrypted using TLS 1.2 or higher.

R036: The system shall be secure against common web vulnerabilities.

R037: The system shall require multi-factor authentication for all administrator accounts.

R038: Administrators shall be able to sign in with only a username and password.

R039: Customer payment card details shall not be stored on the platform's own servers.

8. Performance and Availability

R040: The system shall support at least 5,000 concurrent users without degradation of service.

R041: The system should be fast during normal usage.

R042: The checkout process shall complete within 3 seconds for 95% of transactions under normal operating conditions.

R043: The platform shall maintain 99.9% uptime measured monthly.

R044: The system shall be easy to use.

R045: The system shall scale to handle large traffic volumes during sale events.

9. Reporting and Analytics

R046: The system shall generate a daily sales summary report for administrators.

R047: The system shall allow administrators to export reports.

R048: The system shall track product page views for analytics purposes.